

Project 1: S3 Static Website Hosting

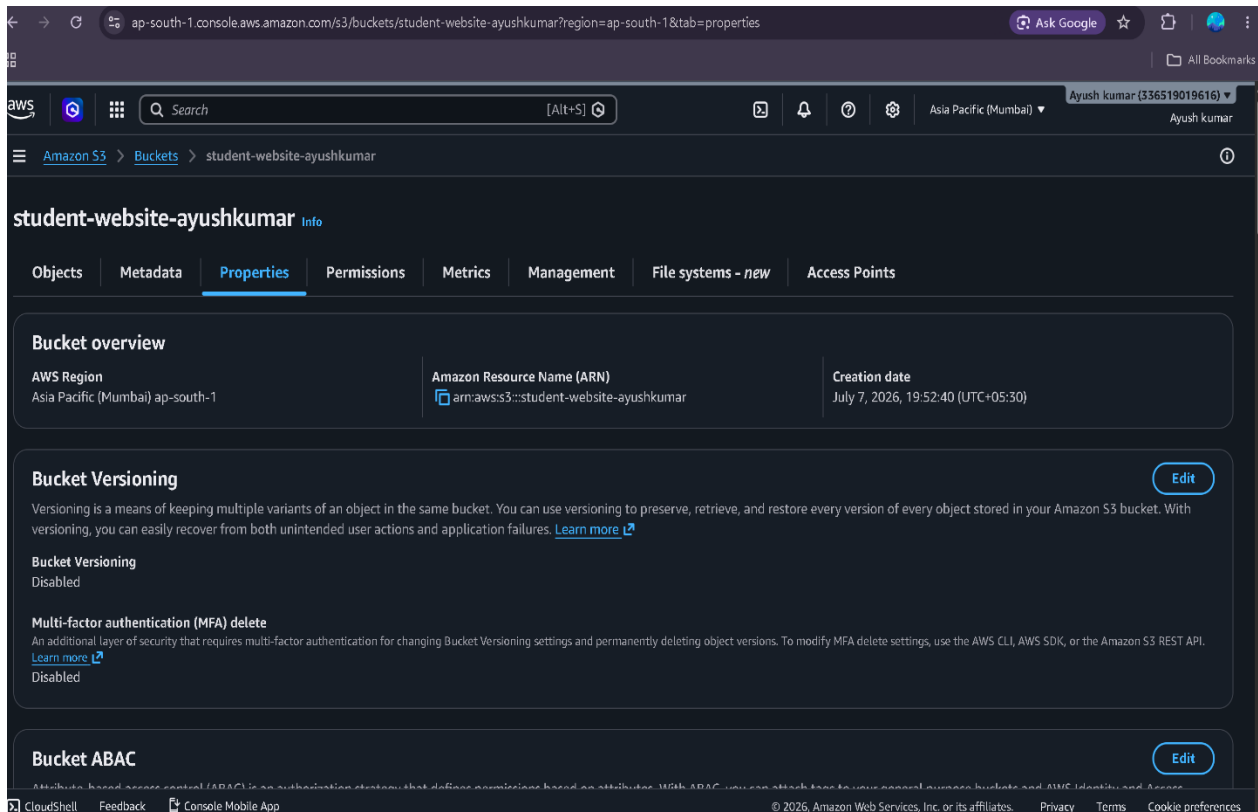
1. Objective

The objective of this project is to successfully host a static HTML website using an Amazon S3 bucket without relying on any underlying web servers.

2. Step-by-Step Implementation

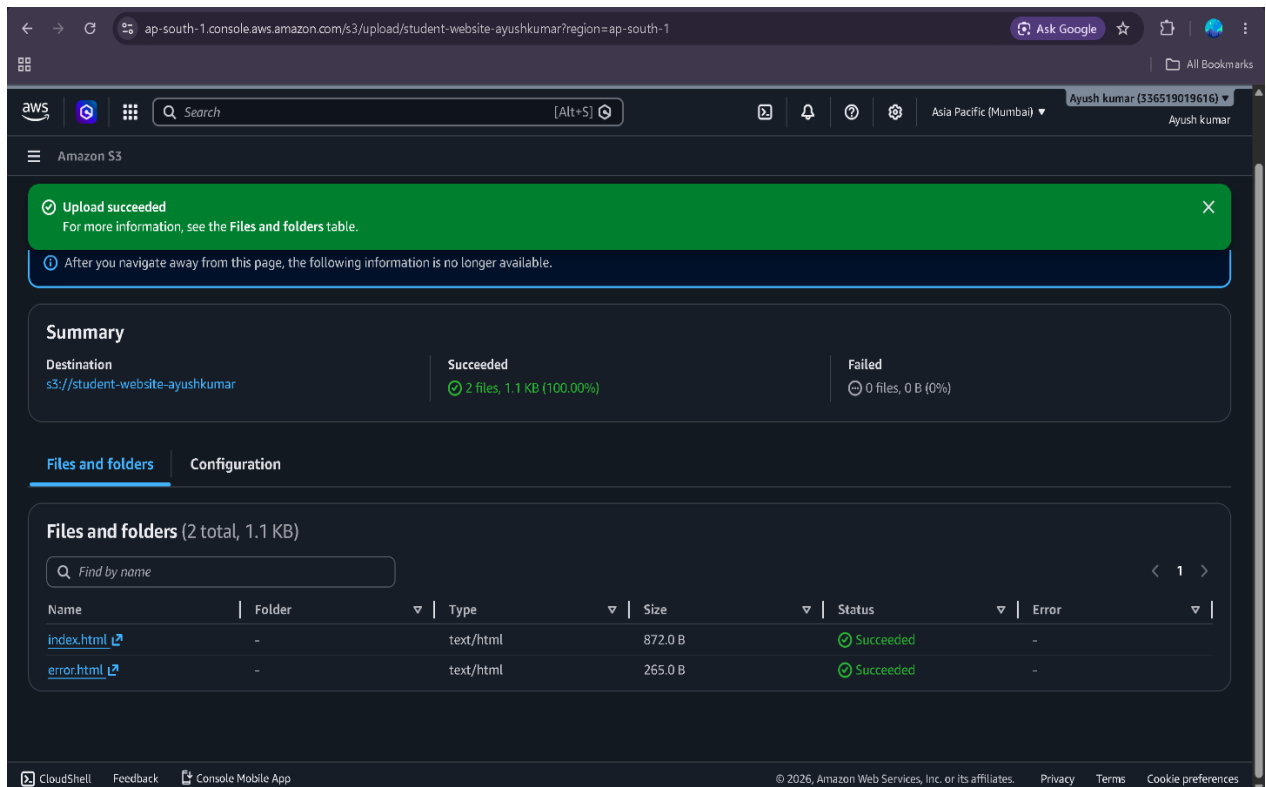
Step 1: Create an Amazon S3 Bucket

1. Log in to the **AWS Management Console** and navigate to the **S3 Console**.
2. Click on the **Create bucket** button.
3. Configure the following specific parameters:
 - **Bucket name:** student-website-ayushkumar
 - **AWS Region:** ap-south-1 (Mumbai).
4. Under the **Block Public Access settings for this bucket** section, **uncheck** the box labeled **"Block all public access"**.
5. Check the acknowledgment box warning that the bucket will become public, then click **Create bucket**.



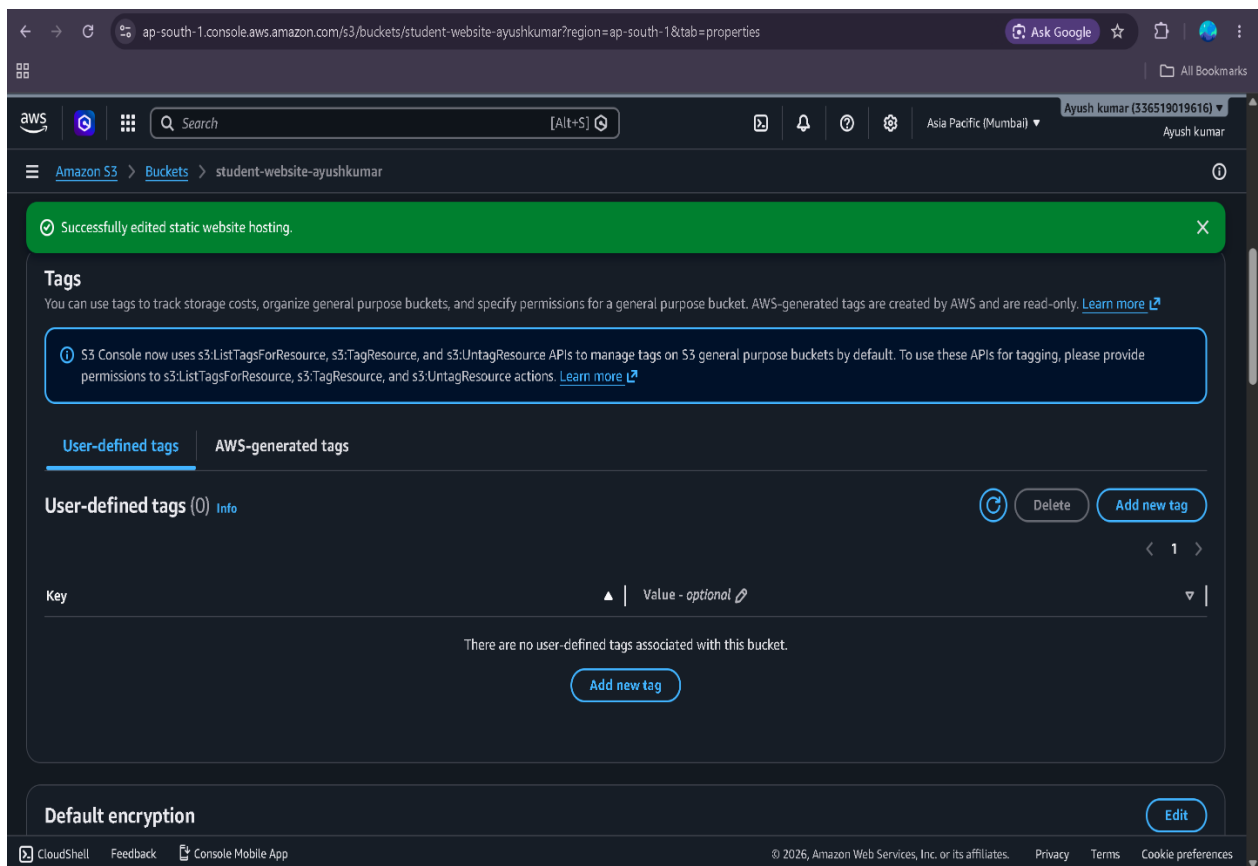
Step 2: Prepare and Upload HTML Files

1. Two local files were created using a text editor:
 - **index.html** (The primary landing page)
 - **error.html** (The custom 404 fallback page)
2. Click on the created bucket name to enter it, and click the **Upload** button.
3. Drag and drop both files into the upload wizard and hit **Upload**.



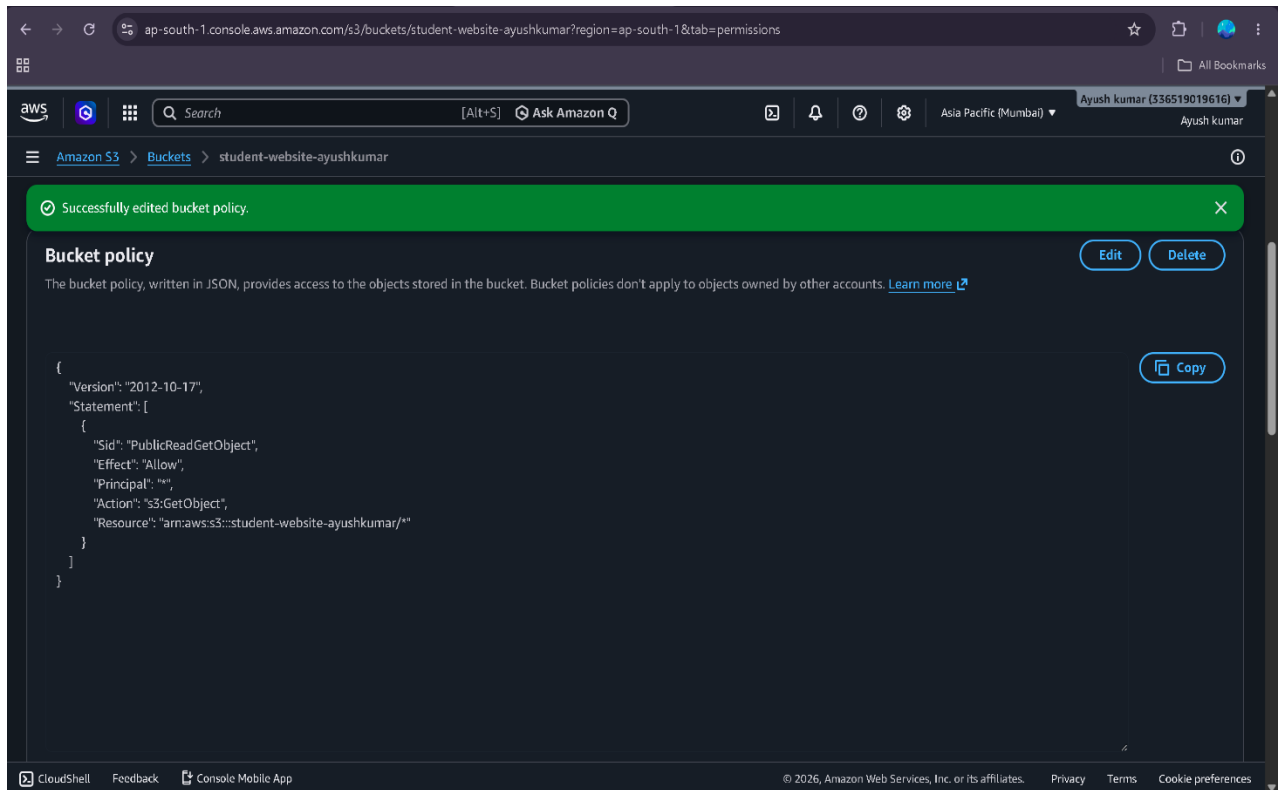
Step 3: Enable Static Website Hosting

1. Select the **Properties** tab from the top menu of your S3 bucket.
2. Scroll to the bottom to find the **Static website hosting** pane and click **Edit**.
3. Select **Enable** and choose **Host a static website**.
4. Provide the exact filenames for the routing components:
 - **Index document:** index.html
 - **Error document:** error.html
5. Click **Save changes**.



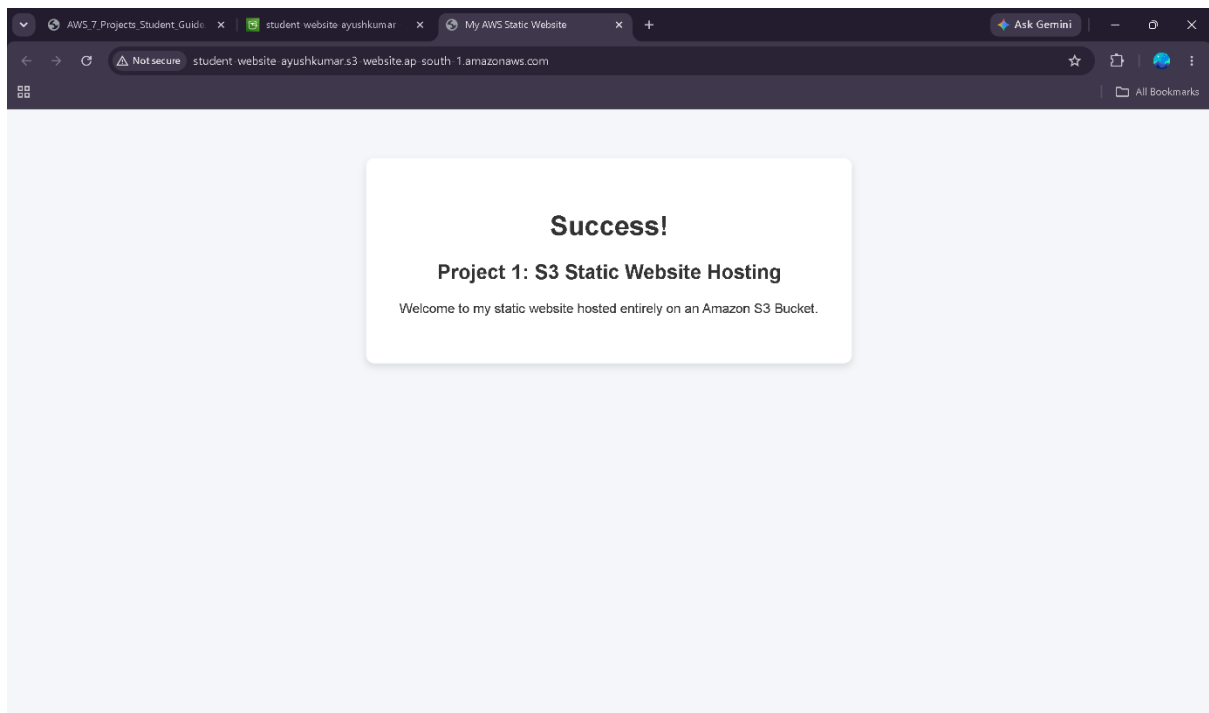
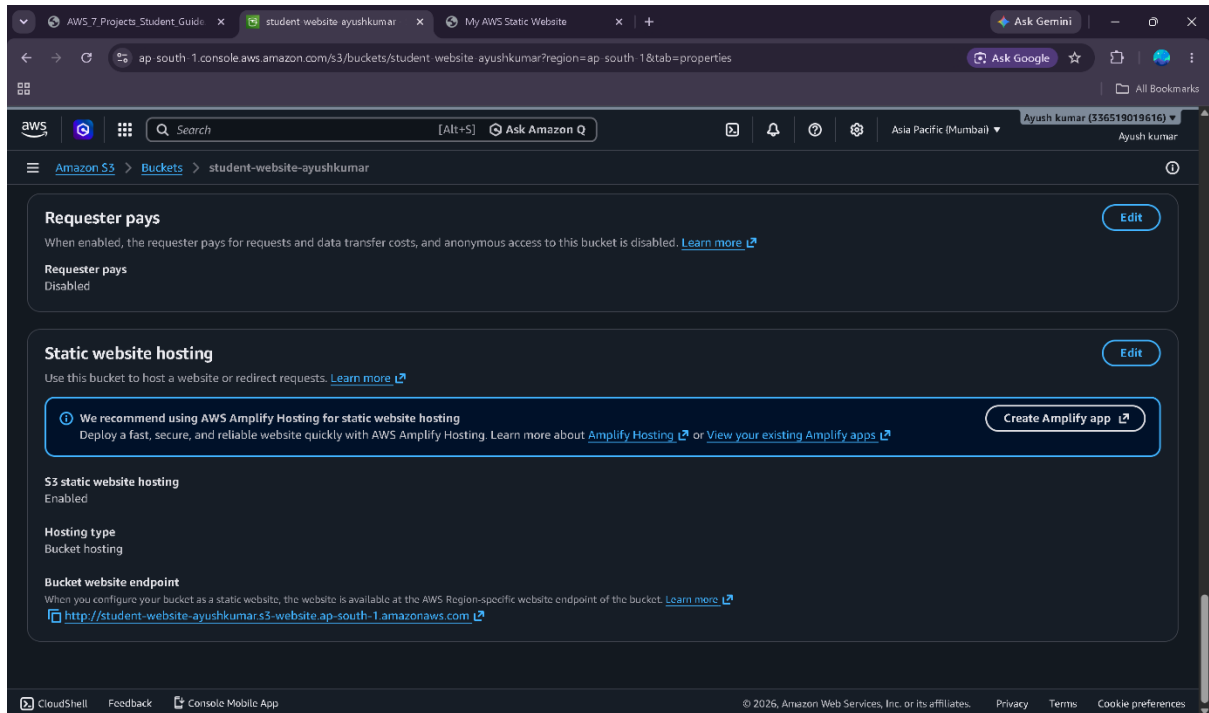
Step 4: Configure Bucket Policy for Public Read Access

1. Switch over to the **Permissions** tab of the bucket.
2. Scroll down to the **Bucket policy** window and click **Edit**.
3. Paste the following structured JSON code block to permit public read access.
4. Click **Save changes**.



Step 5: Live Verification

1. Return to the **Properties** tab and scroll to the bottom.
2. Under the **Static website hosting** section, locate the unique **Bucket website endpoint** URL.
3. Click the generated endpoint link to test it live in a web browser. ([link](#))



3. Conclusion

The S3 bucket was successfully configured to host static web content publicly. By modifying public access configurations, creating an appropriate resource-based access policy, and establishing default indices, serverless static web distribution was completely achieved.